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Service Quality on Online Platforms: Empirical Evidence About Driving Quality at Uber

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Abstract. Online marketplaces have adopted new quality control mechanisms that can accommodate a flexible pool of providers. In the context of ride-hailing, we measure the effectiveness of these mechanisms, which include ratings, incentives, and behavioral nudges. Using telemetry data as an objective measure of quality, we find that drivers not only respond to user preferences but also improve their behavior after receiving warnings about their low ratings. Furthermore, we use data from a randomized experiment to show that informing drivers about their past behavior improves quality, especially for low-performing drivers. Lastly, we find that UberX drivers exhibit behavior comparable to that of UberTaxi drivers, suggesting that Uber's new quality control mechanisms successfully maintain a high level of service quality.

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Keywords: online platforms • ride hailing • reputation systems • nudges • driving quality • telemetry data

1. Introduction

Online marketplaces have transformed many industries. Platforms like Uber, TaskRabbit, and Airbnb now allow small independent providers to sell or rent underutilized goods, work during their free time, and smooth their income during unemployment spans (Katz and Krueger 2016, Chen et al. 2019). When the provider base expands and diversifies and with sufficient competition, potential benefits to consumers include lower prices, more variety, and more flexibility.

Offering goods and services from independent flexible providers also poses challenges for platforms. They need to convince consumers that strangers can provide high quality. The traditional approach of relying on ex ante screening has disadvantages for a model that aims to make use of a flexible pool of independent service providers, many of whom are new to the service or are working part time. New platforms thus follow a streamlined screening process, and to ensure high quality, they instead employ a variety of new technology-enabled methods, harnessing them to shape incentives and provide nudges to providers. For example, they use rating systems extensively, allowing customers to monitor quality through simple interfaces. They combine ratings

with incentive systems that remove individuals who violate quality standards and reward those who do well. Some companies also collect objective real-time measures of quality they can share with their providers to nudge them to perform well.

In this paper, we investigate how well the ex post quality control methods employed by online marketplaces effectively bring about a high service quality, focusing on the context of rides intermediated by Uber. We use a combination of natural experiments and designed randomized experiments to assess the extent to which service providers—drivers—respond to different types of information, incentives, and nudges, shedding light on the mechanisms that induce a higher quality.

The ride-hailing setting that we study is especially well suited for our purposes. Unlike many other markets, where only subjective measures of quality like ratings or reviews are available, in this setting, we observe objective measures of quality in a large sample: Uber collects telemetry metrics such as speed, acceleration, braking, phone handling, and routing. In contrast to subjective measures, our telemetry metrics do not vary endogenously with equilibrium reporting behavior or incentives. This allows us to execute a detailed analysis

of different methods of ex post quality control and their impact on these objective telemetry metrics.

As a building block toward our main results, we first analyze riders' preferences for driving behavior. This allows us to determine what type of behavior should be considered high quality, and, thus, what driving behavior Uber would like to induce. To that end, we use trip rating as a measure of satisfaction. At the end of each trip, riders get the chance to rate the driver on a scale between one and five stars. After controlling for driver, origin, destination, and time of the week, we find that riders give higher ratings to trips with fewer strong brakes and accelerations. They also prefer trips where the driver does not handle the cell phone. These results are consistent with riders preferring safer trips. Riders also give higher ratings when drivers drive at a steady intermediate speed, suggesting that there is some tension between safety and arriving quickly at the destination. Finally, riders prefer shorter trips, both in duration and distance, and they prefer pickups and drop-offs closer to their requested locations. Based on these findings, we build a score that aggregates driving metrics into a one-dimensional measure of driving quality based on a predictive model for the trip rating.

We then move on to the core of our paper, where we quantify the impact of Uber's quality control mechanisms. We first analyze the role of incentives and information provided through ratings. When drivers' ratings fall below certain thresholds, Uber sends them notifications with the aim of improving their behavior, and if ratings keep decreasing, drivers are eventually taken off the platform. We find, first of all, that the drivers that Uber removes—those with persistently low ratings—behave significantly worse than the average driver based on driving metrics and scores. Thus, ratings provide Uber with a simple criterion to remove poorly performing drivers from the platform.

We also find that drivers' quality improves substantially—in terms of both scores and metrics—after receiving notifications about their low ratings: they handle their phones less, accelerate less, drive at steadier speeds, take shorter routes, and pick up and drop off passengers closer to their intended locations. These effects are strong even for the first notification, which occurs far away from the threshold at which they are removed from the platform. Thus, we conclude that, besides direct economic incentives, ratings and notifications work as behavioral nudges that induce better driving behavior. Importantly, the effects of notifications are long-lasting: they persist even after drivers receive notifications from Uber informing them that they are no longer at risk of being taken off the platform once their ratings improve.

Giving drivers feedback about their past behavior also has a positive impact on driving behavior. At the time our data were collected, Uber sent a weekly report

to drivers summarizing how they performed according to telemetry data. Uber conducted a large-scale randomized experiment that introduced a significant upgrade to these reports. Treated drivers gained access to a dashboard within the Uber app where they could analyze their driving behavior for individual trips. We find an improvement in the behavior of treated drivers. Furthermore, the effect is stronger for drivers who performed in the bottom 10th percentile before the experiment started. All these results indicate that Uber's quality control mechanisms positively impact driving behavior. However, it remains possible that, despite these mechanisms, Uber provides poor overall quality due to the lack of ex ante screening. The rest of our paper examines whether that is the case. We take advantage of the fact that, apart from UberX (Uber's main ride-hailing service), Uber also offers a product called UberTaxi that allows riders to request standard taxis licensed by the local government. Whereas the quality provided by UberX is largely controlled through ratings, incentives, and nudges, UberTaxi mostly relies on traditional screening through the licensing process. Differences in driving behavior between both groups cannot be directly interpreted as the effect of different quality control methods because there are several other differences between both products. Drivers belong to different populations with different demographic characteristics. Monetary incentives also differ: UberX drivers are much more likely to own their car, and fare structures are different. Nevertheless, a direct comparison between the quality provided by both products allows us to determine the viability of a ride-hailing service that relies almost entirely on ex post quality control mechanisms with limited ex ante screening.

We find that the overall quality of UberX and UberTaxi trips, as measured by our scores, is roughly equivalent. However, important differences underlie this result. UberX drivers brake and accelerate less, tend to speed less, and pick up and drop off drivers closer to where they want, all of which are desirable behaviors. On the other hand, they handle their phone more and tend to take longer routes. Our scores suggest that riders prefer the way UberX drivers control their vehicle but favor the routing behavior of UberTaxi drivers.¹ On balance, riders are roughly indifferent between the behavior of drivers of both products. These results confirm that Uber's quality control systems are sufficient to ensure a level of quality comparable to that of taxis, despite not relying on ex ante screening.

We also measure heterogeneous differences between UberX and UberTaxi drivers to provide additional evidence that Uber's quality control systems operate as intended. We find that the gap between UberX and UberTaxi is smaller during rush hour, in particular for speed and hard brakes, when riders are more likely to be in a hurry. This is consistent with UberX providing

stronger incentives to cater to riders' preferences, and speeding up at that time. Our findings have important managerial implications. Our main results show that there are several mechanisms platforms can use to ensure a high quality. They can use rating systems coupled with messages to low-rated providers and a deactivation policy, as well as real-time monitoring and feedback to nudge providers to improve their quality. Our results also deliver key messages for regulators concerned about quality. We find that markets that mostly use ex post quality control systems can provide similar quality as traditional markets that rely on ex ante screening and occupational licensing. Regulators should thus consider allowing marketplaces to use well-designed incentives, nudges, and information systems as effective substitutes for rigorous ex ante screening.

1.1. Related Work

In a paper that is closely related to ours, Liu et al. (2021) find that Uber drivers are less likely than taxi drivers to take detours from airport trips. The main implication is that Uber's incentive systems are effective in eliminating this kind of moral hazard. In contrast to their work, we analyze driving behavior, a different type of incentive problem, and we focus on the mechanisms through which incentive systems affect behavior.

Our work is part of a broader literature that analyzes rating and review systems in the digital economy (Dellarocas 2006, Tadelis 2016). Many of these papers focus on the behavior of consumers when they rank providers (Resnick and Zeckhauser 2002, Resnick et al. 2006, Nosko and Tadelis 2015, Proserpio and Zervas 2017, Zervas et al. 2021, Filippas et al. 2022) and on how consumers respond to ratings and rankings, such as in the context of online bookstores (Chevalier and Mayzlin 2006) and restaurants (Luca 2011). In contrast, we focus on provider behavior and how it is influenced by ratings and reviews.²

A broad literature analyzes how nudges and information affect agents' behavior beyond what can be explained by incentives (Thaler and Sunstein 2009, Allcott and Kessler 2019). Two papers focus on driving behavior, with findings that are consistent with some of our results. Jin and Yu (2021) find that text messages can nudge ride-hailing drivers to reduce cell-phone handling, and Choudhary et al. (2022) find that private car drivers respond positively to information on past behavior, especially if they were not good drivers previously. Besides these empirical findings, a key contribution of our paper is to highlight their significance in the context of quality provision in the gig economy.

Several other papers provide evidence that agents react to nudges and information in different contexts, such as surgical performance (Kolstad 2013), energy consumption (Allcott and Rogers 2014), health insurance (Handel 2013), taking medicine (Macharia et al.

1992), voting (Gerber et al. 2003), and charitable donations (Frey and Meier 2004, Shang and Croson 2009, Edwards and List 2014). Our work shows evidence of one setting where nudges, coupled with other mechanisms, are used by companies to achieve an outcome that is desirable to consumers.

A growing literature analyzes several aspects of Uber and ride-hailing markets. Our work most closely relates to those that focus on the labor supply side (Chen and Sheldon 2015, Hall and Krueger 2016, Cook et al. 2020, Hall et al. 2023). A recurring theme is the value of labor flexibility introduced by ride hailing platforms (Chen et al. 2019, Angrist et al. 2021). This kind of flexibility could in principle be accompanied by a reduction in quality; the evidence in our paper establishes that this potential cost is not borne out in practice and documents several mechanisms that contribute to Uber's ability to maintain quality.

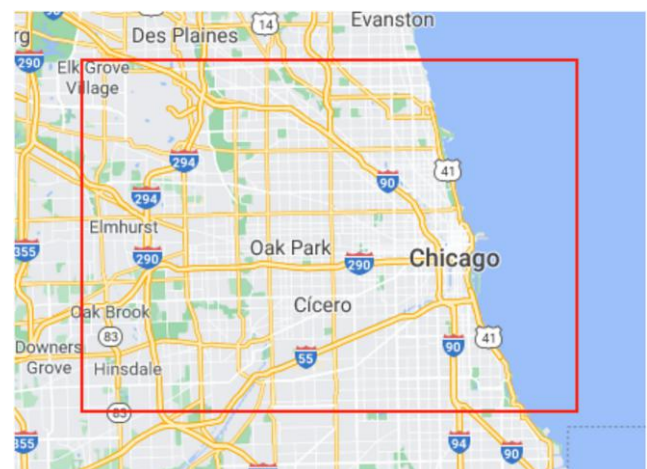
1.2. Roadmap

Section 2 describes the Uber market and the data we use. In preparation for our main analysis, in Section 3, we analyze riders' preferences over driving metrics and construct the driving quality scores that we use as our main outcome variables. Section 4, the core of our analysis, measures how online marketplaces' ex post quality control methods—ratings, incentives, and behavioral nudges—affect driving behavior. In Section 5, we compare the performance of UberX and UberTaxi to provide further evidence on the success of these new quality control mechanisms. Finally, we conclude in Section 6.

2. Uber, Telemetry, and Ratings

We analyze the Uber market in Chicago during the first half of 2017. We limit our analysis to a region around downtown Chicago that is large enough to include

Figure 1. (Color online) Region of Analysis



Note. This figure highlights the region in Chicago we focus on.

Midway and O'Hare, the main airports in the region, but that excludes some far-away suburbs (Figure 1).

Our data involve two Uber products. Our primary analysis focuses on the first product, UberX, which is Uber's main ride-hailing option and its largest product by number of rides. Drivers typically own their cars, and entry requirements are fairly simple: being at least 21 years old; having a driver's license, vehicle registration and insurance; and passing an online screening that reviews driving record and criminal history. The second product is UberTaxi, which matches riders to taxis licensed by the City of Chicago. Taxi drivers must be Chicago public chauffeurs, which involves a lengthy licensing process.³ There are two common ownership models for taxis in Chicago. Some drivers are independent operators and own both the taxi and the medallion. Most other drivers lease both the taxi and the medallion for 12-hour shifts at a flat fee and retain all earnings from working during the shift.

We focus on trips over a several month period in early 2017. The number of UberTaxi trips is much smaller than the number of UberX trips, but this is true for all cities; we chose Chicago as our region of analysis because it is the market with the largest number of UberTaxi trips. Our main data set includes 6,901,200 UberX trips and 139,716 UberTaxi trips after filtering out trips in which any driving metrics, which we describe below, are missing.

2.1. Quality Metrics

Uber collects telemetry data from drivers' smartphones every two seconds while the Uber app is open. The raw data include location, speed, and acceleration.⁴ Our analysis focuses on 10 trip-level metrics, whose distributions are shown in Figure 2. The first two metrics are the *accelerations* metric, the fraction of acceleration events during the trip where acceleration went above 2 m/s^2 , and the *brakes* metric, which is defined similarly.⁵

Uber developed two classifiers based on accelerometer data that predict whether a driver's cell phone was mounted and whether the driver was handling the cell phone (i.e., moving it while holding it with his hands). We define the *handling* and *mounted* metrics as the average of the predictions from these classifiers over the two middle time quartiles of a trip. We do not take into account what happens at the beginning and at the end of a trip because drivers are especially likely to use their phone at the beginning and end of a trip but often in a way that does not necessarily interfere with safety or contribute to passenger dissatisfaction.

Our metrics for speed are based on a model developed by Uber to construct speed metrics for each segment (which roughly corresponds to a block) relative to other cars that went through the same segment. The *contextualized speed* for a segment is the percentile within the distribution of speeds for all cars that went through that segment. We focus on two metrics at the trip level:

Speed low and *speed high*, the 10th and 80th percentiles, respectively, of all contextualized speeds in a trip. They measure how fast the driver was going during the slowest and fastest segments of the trip, relative to other traffic.

We also define four metrics that are related to the route taken by the driver. The first two assess whether the driver chose a good route: our *distance* metric, the log of the ratio of the actual trip distance to the distance Uber estimated before the trip started, and our *duration* metric, the log of the ratio of the actual duration to the duration Uber estimated before the trip. Uber does not estimate distances and durations for UberTaxi trips, so in analyses that include UberTaxi trips, we use a modified version of these metrics that is based on estimated distances and durations that we impute based on a random forest prediction (see Online Appendix B.3). The other two metrics are the *pickup* and *drop-off* metrics, which measure the distance between the requested pickup point and the place where the pickup actually took place, and the distance between the requested drop-off point and the place where the rider was actually dropped off.⁶

There is a large number of alternative metrics we could have used to measure brakes, accelerations, and speed. For instance, we could have used thresholds other than 2 m/s^2 to measure brakes, or we could have used different contextualized speed percentiles. We selected our main variables based on a regularized regression model (Lasso) to predict a trip's rating based on all these variables and their squares. We selected the variables that dropped out last as the penalty increased. The details of this procedure are in Online Appendix B.1. Our main results, however, do not change when we choose alternative metrics.

2.2. Ratings

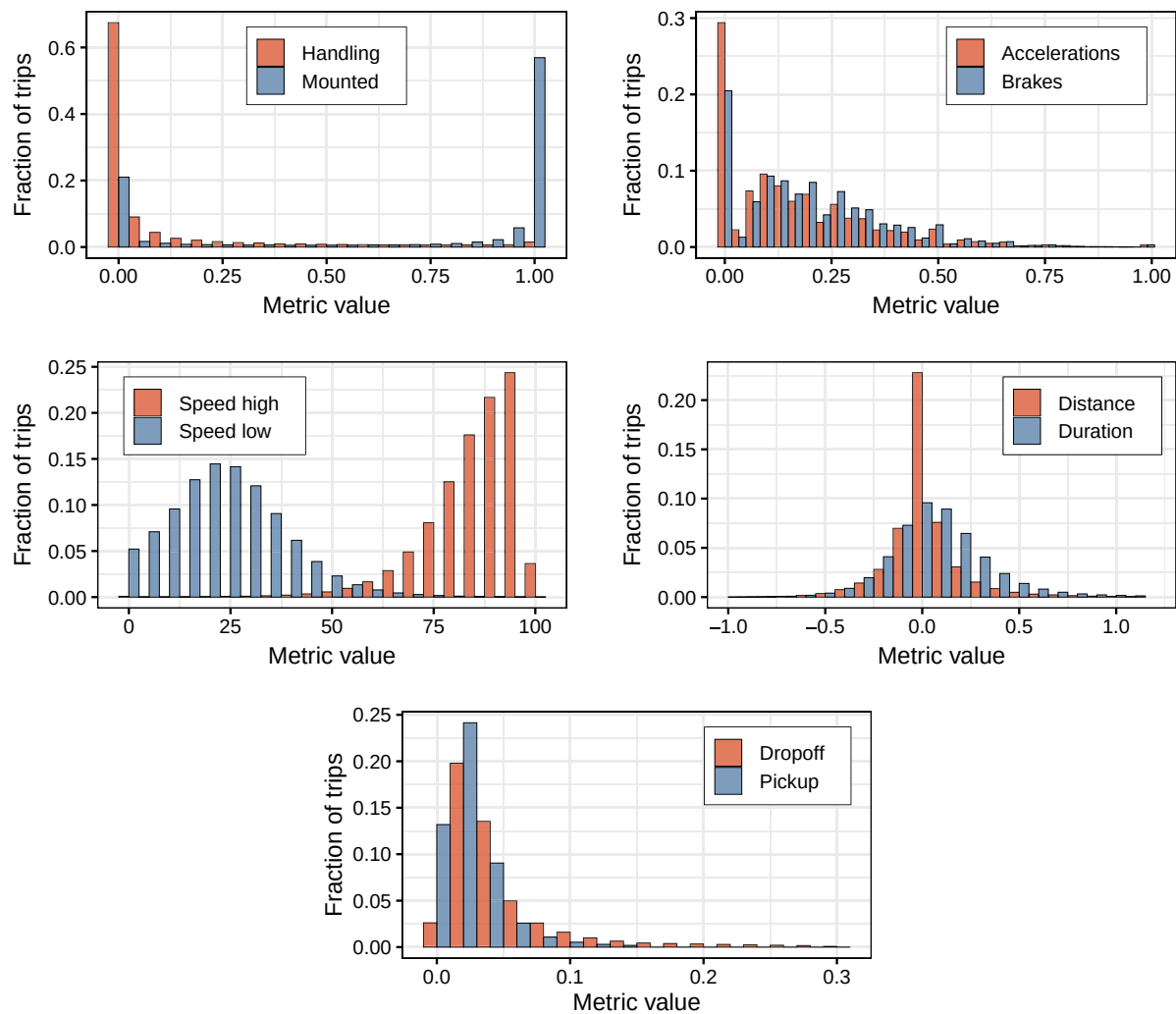
After every trip, the passenger can give a one- to five-star rating to the driver; 28.8% of the trips in our sample were rated. Figure 3(a) shows the distribution of ratings. Most trips receive five stars, consistent with the behavior studied by Filippas et al. (2022). UberTaxi trips tends to get a larger fraction of four-star ratings.

Uber uses the *app rating*, the average of the last 500 trips, as its main measure of driver quality; drivers can see it in the app, and it is shown to passengers upon being matched to a driver before pickup. Figure 3(b) shows its distribution. Uber stops assigning trips to UberX drivers ("deactivation" of a driver) when ratings drop below certain thresholds, but only after a process that involves giving notifications to drivers when their ratings approach the deactivation threshold. We describe the process in detail in Section 4.1.

3. Driving Behavior and Rider Preferences

In this section, we explore what driving behavior riders prefer. This is a key step toward our main results because it allows us to determine what should be

Figure 2. (Color online) Distribution of the Driving Metrics for UberX Trips

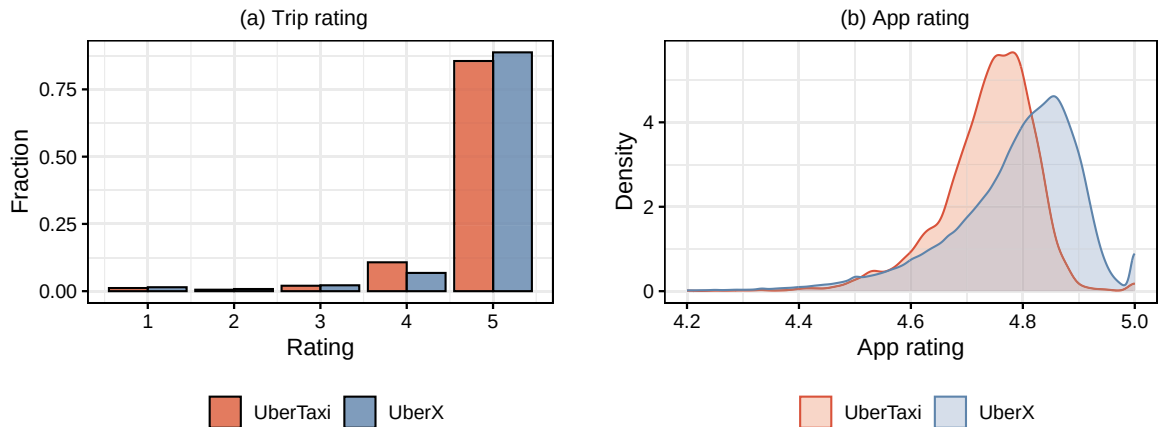


considered high-quality behavior. Based on that analysis, we build one-dimensional scores that summarize driving behavior, and we use them as outcome variables in the rest of the paper.

3.1. Rider Preferences

In order to determine rider preferences for driving behavior, we use different methodologies to predict the rating a rider will give to a trip, based on the observed

Figure 3. (Color online) Distribution of Ratings



Notes. (a) Histograms of the ratings given to individual UberX and UberTaxi trips. (b) Kernel density plots of drivers' app rating (the average of the last 500 trips).

driving metrics. This procedure faces a variety of problems. One challenge is that different kinds of trips (at different times of the day, with different origin and destination) lead to different driving behavior, such as slower trips downtown or during rush hour, some of which might be intrinsically more satisfying for riders. A second challenge arises due to unobserved characteristics that have an effect on satisfaction, such as the driver's personality, which might be correlated with driving behavior.

We address both issues by controlling for the driver and the type of trip to difference out fixed characteristics of the driver that lead to higher satisfaction (such as the comfort of the car) and fixed characteristics of the trip (such as traffic or time of the day).⁷ We thus estimate how *changes* in driving metrics lead to changes in consumer satisfaction, fixing the driver and the type of trip.

To control for the type of trip, we divide our sample into 128 rectangles by origin coordinates and 128 rectangles by destination coordinates. The rectangles are sized so that they have similar numbers of trips. We also divide our sample into 15 hour-of-the-week intervals. This results in 245,760 buckets as the cartesian product of all origin, destination, and hour of the week divisions. Most of our regressions include fixed effects based on these groups, which we call *trip characteristics* fixed effects. Online Appendix A gives further details about these trip characteristic groups, and it shows that, despite the large number of groups, most trips in our sample are in groups with more than five trips.

3.1.1. Individual Metrics. We start with simple linear regressions to understand how ratings relate to individual metrics. We estimate regressions of the following form:

$$y_i = \beta m_i + \gamma X_i + \epsilon_i. \quad (1)$$

Trips are indexed by i . The left-hand-side variable y_i is a measure of trip rating, and m_i is a vector of driving metrics; X_i is a set of fixed effect dummies. Table 1 shows estimates of this equation for our sample of UberX trips. All driving metrics are normalized, so coefficients measure how ratings change if metrics change by one standard deviation.

Columns (1) and (2) show results for regressions in which the dependent variable is the five-star rating, and in columns (3) and (4), the dependent variable is an indicator that takes the value of one when the rating is equal to five. Riders dislike phone handling and hard brakes. Some specifications show a weak preference for cell phones being mounted. When we focus on regressions with driver fixed effects, we see that riders dislike hard accelerations. The four metrics we have mentioned reflect preferences for safer trips; these trips may also provide a more comfortable ride. Preferences for speed metrics, on the other hand, are more nuanced. Riders

prefer higher low speeds and lower high speeds, that is, trips in which drivers stay at an intermediate speed throughout the trip. This reflects a compromise between a safe smooth ride and getting quickly to the destination. Riders have strong preferences for shorter trips, both in terms of distance and duration. They also prefer to be picked up and dropped off close to the requested locations.

The dependent variable in columns (5) and (6) is a dummy for whether the trip was rated. Coefficients tend to have opposite signs relative to previous columns (except for speed high and drop-off). This suggests that riders are more likely to rate trips when they are unsatisfied. This type of bias is one motivation for our approach described below; by creating a score for the quality of the ride that can be evaluated whether the ride was actually rated, we avoid the challenge of dealing with non-random missing ratings. Thus, when interpreting magnitudes, we caution that our predictions of ratings only apply to the type of ride that was actually rated in practice rather than overall satisfaction if all rides were rated.

One potential concern with Table 1 is that traffic may be correlated with certain forms of driving, and riders blame drivers for bad traffic conditions. That would result in a correlation between driving metrics and ratings that is unrelated to rider preferences for those metrics. In Online Appendix D.2, we construct a measure of traffic conditions and show that our results are robust to controlling for traffic. A second concern is that the trip characteristic fixed effects we define may be too coarse spatially. Online Appendix D.3 shows that our results are robust to defining trip characteristic fixed effects in different ways.

3.1.2. Flexible Functional Form. We now follow a more flexible approach to capture the dependence of rating as a function of driving metrics. We estimate

$$y_i = \mu_{d(i)} + \nu_{c(i)} + s(m_i; \theta) + \epsilon_i, \quad (2)$$

where $d(i)$ indexes the driver and $c(i)$ indexes the trip characteristics group. Thus, $\mu_{d(i)}$ and $\nu_{c(i)}$ represent driver and trip characteristics fixed effects.

The term $s(m_i; \theta)$ is a flexible function of driving metrics. It is the sum of three high-order polynomials. The first one is the interaction of a quadratic function of handling and a quadratic function of mounting. The second one is the interaction of a quadratic of brakes, a quadratic of accelerations, a quartic of high speed, and a quartic of low speed. The third term is the interaction of a cubic of distance, a cubic of duration, a quadratic of pick-up, and a quadratic of drop-off.⁸ To avoid overfitting (this specification for $s(m_i; \theta)$ has 378 parameters), we regularize our model with a lasso penalty, where higher-order terms have higher penalties. We do not

Table 1. Rating Response to Driving Metrics

Variable	Dependent variable					
	Rating		Rating is five		Rated	
	(1)	(2)	(3)	(4)	(5)	(6)
Mounted	0.0096*** (0.0009)	0.0015 (0.0013)	0.0045*** (0.0005)	0.0010 (0.0006)	0.00002 (0.0002)	−0.0011** (0.0005)
Handling	−0.0018** (0.0008)	−0.0050*** (0.0009)	−0.0005 (0.0004)	−0.0016*** (0.0004)	0.0006** (0.0002)	−0.0002 (0.0003)
Brakes	−0.0089*** (0.0007)	−0.0032*** (0.0006)	−0.0041*** (0.0003)	−0.0014*** (0.0003)	0.0009*** (0.0002)	0.0014*** (0.0002)
Accelerations	0.0014** (0.0007)	−0.0035*** (0.0006)	0.0009*** (0.0003)	−0.0015*** (0.0003)	0.0015*** (0.0002)	0.0013*** (0.0002)
Speed low	0.0069*** (0.0007)	0.0031*** (0.0006)	0.0033*** (0.0003)	0.0015*** (0.0003)	−0.0026*** (0.0002)	−0.0028*** (0.0002)
Speed high	−0.0019*** (0.0007)	−0.0065*** (0.0006)	−0.0014*** (0.0004)	−0.0036*** (0.0003)	−0.0020*** (0.0002)	−0.0026*** (0.0002)
Distance	−0.0156*** (0.0008)	−0.0155*** (0.0007)	−0.0046*** (0.0004)	−0.0047*** (0.0003)	0.0006*** (0.0002)	0.0005** (0.0002)
Duration	−0.0265*** (0.0008)	−0.0232*** (0.0007)	−0.0103*** (0.0004)	−0.0089*** (0.0003)	0.0066*** (0.0002)	0.0059*** (0.0002)
Pickup	−0.0133*** (0.0007)	−0.0115*** (0.0006)	−0.0065*** (0.0003)	−0.0058*** (0.0003)	0.0082*** (0.0002)	0.0080*** (0.0002)
Dropoff	−0.0158*** (0.0008)	−0.0139*** (0.0006)	−0.0066*** (0.0004)	−0.0057*** (0.0003)	−0.0008*** (0.0002)	−0.0008*** (0.0002)
Trip characteristics fixed effects	✓	✓	✓	✓	✓	✓
Driver fixed effects		✓		✓		✓
Observations	1,991,742	1,991,742	1,991,742	1,991,742	6,901,200	6,901,200

Notes. This table shows results of regressions of rating variables—five-star rating, a dummy for the rating being five, and a dummy for the trips being rated—on driving metrics. All metrics are normalized to mean zero and variance one. Standard errors are clustered by driver.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

penalize fixed effects. We choose penalties by cross-validation (see Online Appendix B.2). Our final specification, which has lower out-of-sample mean squared error (MSE) than simple lasso, is a post-Lasso linear regression that only keeps those terms with a nonzero coefficient from the original lasso regression.

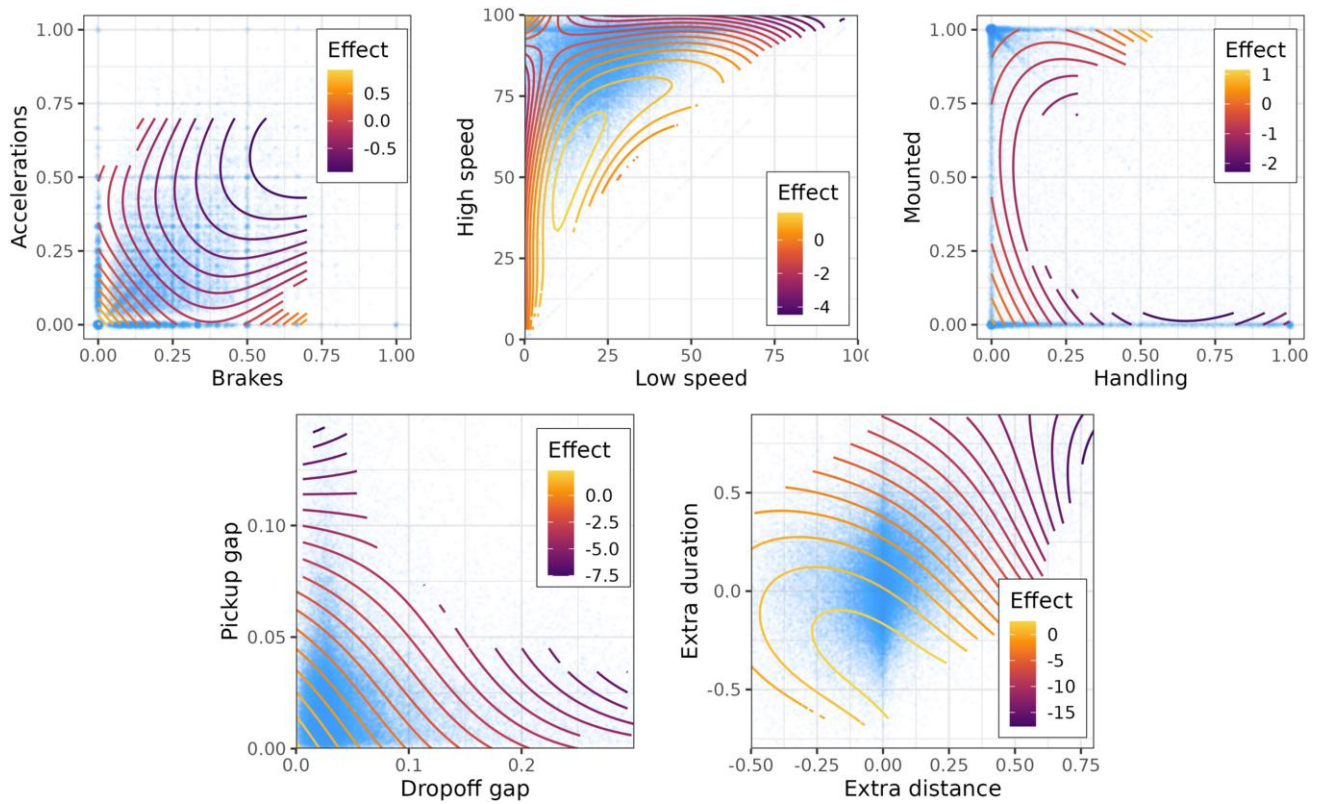
Figure 4 shows the functional form of our estimated $s(m_i; \hat{\theta})$. In each subfigure, we vary two of the metrics. At each point in these plots, we compute the average value of $s(m_i; \hat{\theta})$ over the distribution of the metrics we are not varying. The contours, as well as their color, represent the average value of $s(m_i; \hat{\theta})$. The scatterplot represents the distribution of the two metrics we are varying.

The top left panel confirms that riders prefer fewer hard accelerations and hard brakes. The top middle panel shows preferences for speed. Riders have a bliss point near (20,55). This confirms that riders prefer trips that are neither too fast nor too slow.⁹ In the top right panel, most of the mass is on the top left corner. Around that corner, riders prefer drivers to mount their phone and not to handle it. The bottom left panel shows that riders prefer to be picked up and dropped off close to the requested locations. Finally, the bottom right panel shows that, for the most part, riders prefer shorter and quicker trips, although their preferences flatten out when trips become too short or too quick.

3.2. Driving Scores

In the rest of this paper, we compare the driving behavior across different settings. Although we can use individual driving metrics as outcomes, our previous analysis shows that riders' preferences over these metrics are nonlinear, and, in some cases, nonmonotonic. In order to summarize differences in quality across all metrics, we define a driving score $s_i^F = s(m_i; \hat{\theta})$ based on Equation (2) that we call our *full score* or *score F* (because it is computed from our full model). We measure the score in hundredths of a star. Using our score as our main outcome variable instead of using ratings directly allows us to narrow down on a specific dimension of quality, driving behavior. This allows us to focus on objective behavior that we can observe through telemetry metrics, rather than on subjective preferences captured by ratings—which, as we show in Section 3.3, includes a large amount of rider-specific information that has nothing to do with driver behavior. It also has the advantage that it allows us to compare drivers' behavior when riders do not rate trips, which is the case for more than two thirds of the trips in our sample.

In some specifications, we want to distinguish preferences related to different types of metrics. We thus define a vehicle *control score* or *score C*, denoted by s_i^C . We compute it from a model of the form of Equation (2), but where $s(m_i; \theta)$ only includes terms related to

Figure 4. (Color online) Effect of Driving Metrics on Ratings

Notes. Estimated effect of quality metrics on ratings. Each subfigure shows how the effect changes as two of the metrics change. Contour colors represent the magnitude of the effect. Effects are the average of the effect over the distribution of the rest of the metrics. The dots are a scatterplot to show the distribution of the two variables of interest. Contours are only shown in areas with a significant density.

handling, mounting, acceleration, brakes, and speed, which measure how the driver controls the vehicle. We also define a *routing score* or *score R* , denoted by s_i^R , where $s(m_i; \theta)$ only includes terms related to distance, duration, pickup, and drop-off. Finally, we define a *speed score* or *score S* , denoted by s_i^S , where $s(m_i; \theta)$ only includes terms related to speed low and speed high. The speed score thus captures a subset of the information captured by the control score.

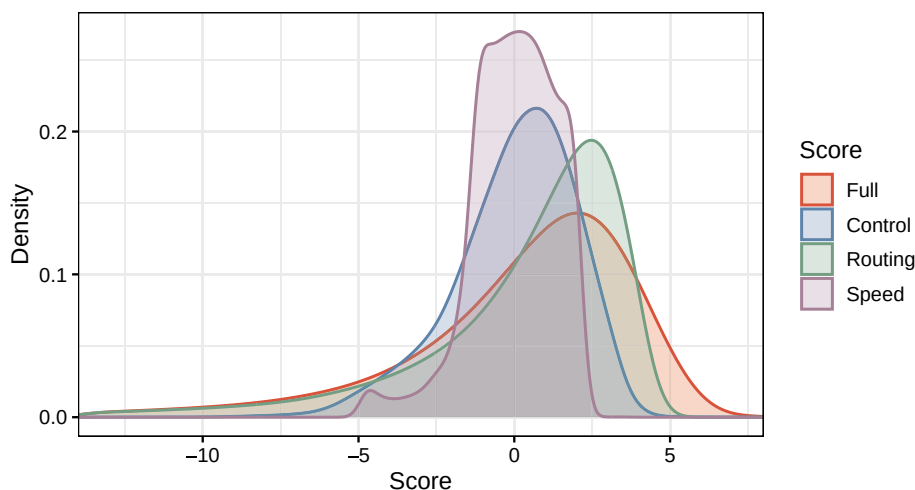
Figure 5 shows the density of our four scores. The full score has the largest variance because it captures variation arising from all metrics. The control and routing scores each account for a substantial fraction of the variance. All four scores have a small standard deviation—4.39 hundredths of a star for the full score, 2.01 for the control score, 4.29 for the routing score, and 1.35 for the speed score. Two factors contribute to this finding. First, the standard deviation of the rating is pretty low because most trips have a rating of five. Second, as we will show in Section 3.3, of all the variation in ratings, only a small fraction is under the driver's control. The magnitude of our scores is on the order of magnitude of the score changes that drivers can achieve.

We construct our scores based on the sample of UberX trips only. Thus, they capture the preferences of

UberX riders, who may have chosen UberX precisely because their preferences tilt toward the driving behavior of UberX drivers. This may be a concern for our analysis from Section 5, where we compare the behavior of UberX and UberTaxi drivers. Online Appendix B.4 shows that those results look very similar when we construct scores based on the sample of all UberX and UberTaxi trips. We also run similar exercises using a score that was constructed from UberTaxi trips only—although we have less confidence in these scores because of the limited size of our sample of UberTaxi trips. See Online Appendix B.4 for a fuller analysis.

3.2.1. Scores as Surrogates. In using scores as an outcome variable, we follow Athey et al. (2019). They propose constructing a *surrogate index*, an outcome variable in settings where the true outcome of interest is missing—because it is a long term outcome, for instance, or because it is not systematically available. The surrogate index is defined as the predicted value of the outcome conditional on a set of intermediate outcomes, the *surrogates*. It can be estimated in a data set that differs from the one used to evaluate the impact of a treatment. The surrogates in our case are the driving metrics, the outcome that is sometimes missing is the

Figure 5. (Color online) Distribution of Scores



Notes. Kernel density plots of the full score, the control score, the routing score and the speed score. The scores are measured in hundredths of a star, and they are centered so that they have mean zero.

trip rating, and the surrogate index are the scores. Thus, we use the score as a way to aggregate telemetry measures in a single dimension that relates to quality as experienced by riders.

We follow Athey et al. (2019) most closely in Section 4.3, where we use the scores to evaluate the impact of a randomized treatment that provides information to UberX drivers about past driving behavior. Athey et al. (2019) consider two assumptions: (i) the surrogates capture the effect of the treatment on the final outcome, and (ii) the relationship between the surrogates and the final outcome does not depend on the treatment. Given that we are focusing our attention on the impact of differences in telemetry metrics on rider satisfaction, it is natural to focus on differences in trip ratings that are captured by the driving metrics, so condition (i) is satisfied by assumption. Condition (ii) is plausible because our experiment affects the information available to drivers but does not directly impact riders.

Athey et al. (2019) show that two results hold under assumptions (i) and (ii). First, the estimated impact of the treatment on the score gives a consistent estimate of the average treatment effect on the final outcome. Second, it can be more efficient to analyze the impact of the treatment on the surrogate index rather than directly on the final outcome (even if that outcome were observed).¹⁰

The rest of our analysis in Section 4 uses scores to evaluate the impact of other treatments (such as warnings and notifications) that are not assigned in a random experiment. The above results hold under the additional assumption of unconfoundedness conditional on controls—the usual assumption that is required in quasi-experimental settings. Finally, in Section 5, we use the score to compare UberX to UberTaxi. Our goal is to

quantify the changes in quality that would occur if riders were to request an UberX trip instead of an UberTaxi trip. We do not argue that we capture the well-being of UberTaxi riders because they may have different preferences. Instead, our scores should be interpreted as differences in how an UberX rider would evaluate the driving behavior.

3.3. What Determines Driving Behavior?

Our results from Section 3.1 give us a clear picture of what kind of behavior riders like, and Section 3.2 give us a way of measuring how well behavior complies with what riders like. However, it is still unclear whether that behavior is determined by the driver, by the rider, or by the type of trip. In this section we explore these questions.

3.3.1. Variance Decomposition. We start by decomposing the variance of ratings, driving metrics, and scores into different components. For outcome y_i , we run

$$y_i = \mu_{d(i)} + \nu_{c(i)} + \gamma_{r(i)} + \epsilon_i, \quad (3)$$

where $r(i)$, $c(i)$, and $d(i)$ represent, respectively, the rider, trip characteristics, and driver of trip i . We thus decompose y_i into a driver effect, a trip characteristics effect, a rider effect, and a residual.¹¹

Table 2 shows the variance of each one of the four terms in Equation (3), where the outcomes y_i are driving metrics, scores, and trip rating. For control metrics and scores, the driver accounts for a significant share of the variation. This is especially true for mounted and handling. The rider is not responsible for much of the variation. We also see that trip characteristics are especially

important for speed and for the pickup and drop-off metrics.¹²

Driver effects account for a small fraction of the variation in ratings. This explains, in part, why our scores have such a small standard deviation. Rider effects, on the other hand, account for a substantial fraction of the variation in ratings. This highlights the limitations of the rating itself as opposed to the scores based on telemetry data we use to evaluate ride quality; our scores are applied systematically to the telemetry measures from each ride, whereas the ratings have rider-specific noise that is unrelated to the driver's performance.

4. Ratings, Incentives, and Nudges

This section presents evidence that the mechanisms Uber uses to encourage high quality positively impact driving behavior and trip quality. We first show that ratings and related notifications nudge drivers into better driving behavior and that when Uber removes drivers with low ratings, it tends to remove those with low-quality driving. We then show that providing feedback to drivers about their driving behavior leads to improvements in driving quality.

4.1. Response to Ratings and Notifications

We now show evidence that drivers respond directly to ratings and to the notifications they receive when ratings become low.

4.1.1. Uber's Rating System. The main element of Uber's incentive system are rules under which UberX drivers with low ratings stop being matched to riders. These rules are coupled with notifications that are sent to drivers when their ratings reach certain thresholds.

Table 2. Variance Decomposition

	Driver	Rider	Trip characteristics	Residual
<i>Mounted</i>	0.830	0.005	0.003	0.164
<i>Handling</i>	0.591	0.012	0.008	0.386
<i>Brakes</i>	0.220	0.024	0.038	0.725
<i>Accelerations</i>	0.333	0.021	0.026	0.628
<i>Speed low</i>	0.098	0.045	0.083	0.787
<i>Speed high</i>	0.137	0.047	0.116	0.651
<i>Distance</i>	0.018	0.058	0.040	0.847
<i>Duration</i>	0.039	0.059	0.042	0.831
<i>Pickup</i>	0.016	0.065	0.075	0.711
<i>Dropoff</i>	0.014	0.076	0.138	0.601
<i>Score F</i>	0.079	0.097	0.091	1.209
<i>Score C</i>	0.284	0.030	0.034	0.653
<i>Score R</i>	0.042	0.100	0.103	1.319
<i>Score S</i>	0.088	0.039	0.051	0.839
<i>Rating</i>	0.046	0.301	0.036	0.627

Notes. The table shows the fraction of the variation of the variable in each row that can be explained by driver fixed effects, rider fixed effects, and trip characteristics fixed effects. The numbers are the result of regressions on three way fixed effects. We limit our sample to drivers and riders that have more than 20 trips in our sample.

Table 3 summarizes the deactivation process, which moves sequentially through the steps in each row. In order to move to the next step, a driver has to satisfy the conditions on average ratings for the last 50 and 500 trips and the condition for the number of rated trips (they cannot enter the process before they complete 500 trips in total). At each one of the steps, the driver gets a notification by email, by text messaging, and through the Uber app. The notification explains that they are getting closer to deactivation and provides links to resources with help to improve ratings. Drivers can also progress backward through the deactivation process if their ratings improve sufficiently. Drivers go back to the state right before the last notification if the app rating goes above certain thresholds.¹³

An important fraction of drivers satisfy the conditions to be in the deactivation process. Sixty-four percent of trips are completed by drivers who have completed more than 500 trips; 7.4% of trips are completed by drivers with a rating below 4.6; and 1.3% are completed by drivers with a rating below 4.4. Even more importantly, a larger fraction of drivers are close to these thresholds and should therefore be worried about their behavior if they want to avoid entering the process or being deactivated. For 11.8% of drivers, receiving a three-star rating for the next 15 trips would result in an app rating below 4.6; for 1.8% of drivers, this sequence would result an app rating below 4.4. Online Appendix E shows further statistics related to the deactivation process.

4.1.2. Response to Rating and Notifications. We first explore how drivers' current app ratings affects the way they drive. Additionally, we want to see if the notification system influences their behavior. For outcome y_i , we estimate the model

$$y_i = \mu_{d(i)} + v_{c(i)} + \alpha r_i^{\text{app}} + \beta w_i + P_2(n_i; \gamma) + \epsilon_i, \quad (4)$$

where r_i^{app} is the rating the driver observes in the app at the time of the trip, and w_i is a vector of dummies that characterizes the stage along the deactivation process in which the driver is in. We include fixed effects at the driver and trip characteristics level.

Because we are including driver fixed effects, we focus on variation within drivers, a large fraction of which might be driven by the experience of the driver. This might be problematic, especially when trying to measure α , the response to the app rating. For that reason, we control for $P_2(n_i; \gamma)$, a quadratic of the number of Uber trips the driver has completed, which is a proxy for experience. In order to identify β we exploit variation due to drivers that crossed some threshold and got a notification. Online Appendix E shows that there is a large number of such crossings, both from above and below.

Table 3. Deactivation Process for UberX

Event	Last 500 rating	Last 50 rating	Rated trips
Notification 1	<4.6	<4.6	25 since first trip
Notification 2	<4.5	<4.5	25 since notification 1
Temporary deactivation	<4.4	<4.4	25 since notification 2
Reactivation		Passed quality improvement course	
Notification 3	<4.4	<4.4	25 since reactivation
Permanent deactivation	<4.4	<4.4	25 since notification 3

Table 4 shows the results of this exercise. In panel A, w_i is simply a dummy for being in any notification state. In other words, whenever a driver gets his first notification, w_i switches from zero to one and stays like that forever. We see that notifications have a positive effect on ratings and on all scores. Table 5, which presents the same exercise for individual metrics, shows that notifications also affect every driving metric in the direction that riders prefer according to Table 1, although not all coefficients are significant. One potential concern with these results is that they may be driven by reversion to the mean. Online Appendix D.7 shows that results are similar if we exclude the last 10 or 20 trips before the driver rating crosses the threshold to receive a notification, suggesting that this form of mean reversion is not the main driver of these findings.

The estimates of the app rating coefficient α suggest that drivers respond to low ratings by changing their behavior in a way that increases future ratings. This can be by improving how they drive, but it can also be through some channels we are not able to measure. For instance, drivers can be more friendly with their riders, or they can start cleaning their car. Columns (2)–(5) suggest that this is mostly unrelated to driving behavior, since we find small, positive coefficients that are marginally significant. A potential concern with giving a causal interpretation to this coefficient is that there may be factors that lead to serial correlation in driver behavior. In Online Appendix D.6, we consider a related exercise in which we measure how different outcomes respond to shocks in the rating from the last trip that the driver completed, using an instrumental variables

Table 4. Response of Scores to Ratings and Notifications

	Dependent variable				
	Rating (1)	Score F (2)	Score C (3)	Score R (4)	Score S (5)
Panel A: General effect of warnings					
Has received notification	0.108*** (0.007)	0.103*** (0.020)	0.051*** (0.017)	0.078*** (0.017)	0.020*** (0.007)
App rating	−0.296*** (0.013)	0.070* (0.036)	0.043* (0.025)	0.065** (0.033)	0.025** (0.012)
Observations	1,983,507	6,874,960	6,874,960	6,874,960	6,874,960
Panel B: Decomposition of effect of warnings					
First notification	0.100*** (0.007)	0.104*** (0.021)	0.049*** (0.017)	0.071*** (0.018)	0.017** (0.007)
Second notification	0.125*** (0.013)	0.060* (0.036)	0.033 (0.028)	0.057* (0.031)	0.003 (0.012)
Third notification	0.239*** (0.036)	0.181* (0.095)	0.232*** (0.081)	0.028 (0.081)	0.081** (0.032)
First notification expired	0.104*** (0.008)	0.106*** (0.025)	0.039** (0.020)	0.108*** (0.021)	0.019** (0.009)
Second notification expired	0.089*** (0.024)	0.025 (0.073)	−0.028 (0.051)	0.063 (0.064)	0.034 (0.023)
Third notification expired	0.277*** (0.053)	0.307* (0.157)	0.405** (0.158)	0.043 (0.138)	0.196*** (0.051)
App rating	−0.298*** (0.013)	0.065* (0.036)	0.040 (0.025)	0.060* (0.033)	0.022* (0.012)
Observations	1,983,507	6,874,960	6,874,960	6,874,960	6,874,960

Notes. This table shows regressions of ratings and scores on app ratings and dummies for having received notifications. Panel A uses a single dummy for having received notifications. Panel B uses different dummies for different stages in the notification process. All regressions include driver and trip characteristics fixed effects and control for a quadratic function of the number of Uber trips the driver has completed. Standard errors are clustered by driver. All driving metrics are normalized to mean of zero and variance of one. A few observations are missing due to drivers that have never been rated.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 5. Response of Metrics to Ratings and Notifications

	Dependent variable									
	Mounted (1)	Handling (2)	Brakes (3)	Accelerations (4)	Speed low (5)	Speed high (6)	Distance (7)	Duration (8)	Pickup (9)	Dropoff (10)
Panel A: General effect of warnings										
<i>Has received notification</i>	0.051*** (0.012)	−0.042*** (0.012)	−0.003 (0.007)	−0.016** (0.007)	0.012** (0.005)	−0.002 (0.006)	−0.013*** (0.004)	−0.014*** (0.004)	−0.015*** (0.004)	−0.002 (0.004)
<i>App rating</i>	0.041** (0.017)	−0.025 (0.019)	0.010 (0.011)	0.002 (0.012)	0.043*** (0.009)	0.017* (0.009)	0.010 (0.007)	−0.023*** (0.007)	−0.007 (0.007)	−0.011* (0.006)
Observations	6,874,960	6,874,960	6,874,960	6,874,960	6,874,960	6,874,960	6,874,960	6,874,960	6,874,960	6,874,960
Panel B: Decomposition of effect of warnings										
1st notification	0.049*** (0.012)	−0.039*** (0.012)	−0.006 (0.007)	−0.017** (0.008)	0.007 (0.006)	−0.012** (0.006)	−0.012*** (0.004)	−0.011*** (0.004)	−0.010** (0.005)	−0.004 (0.004)
2nd notification	0.024 (0.020)	−0.039** (0.019)	−0.012 (0.012)	−0.027** (0.012)	0.005 (0.009)	0.016* (0.010)	−0.013* (0.007)	−0.012* (0.007)	−0.022*** (0.008)	0.009 (0.006)
3rd notification	0.078* (0.047)	−0.117* (0.065)	−0.064** (0.029)	−0.029 (0.029)	0.020 (0.024)	−0.042 (0.026)	−0.025 (0.018)	−0.005 (0.020)	−0.036** (0.018)	0.006 (0.018)
1st notif. expired	0.065*** (0.015)	−0.042*** (0.014)	0.016* (0.009)	−0.004 (0.010)	0.022*** (0.007)	0.017** (0.007)	−0.013*** (0.005)	−0.020*** (0.005)	−0.020*** (0.005)	−0.003 (0.004)
2nd notif. expired	0.009 (0.032)	−0.010 (0.031)	0.016 (0.020)	0.012 (0.019)	0.030* (0.016)	0.033* (0.020)	−0.019 (0.013)	−0.015 (0.015)	−0.013 (0.014)	−0.003 (0.012)
3rd notif. expired	0.109 (0.077)	−0.176 (0.123)	−0.098** (0.042)	−0.132** (0.059)	0.078** (0.034)	−0.050 (0.032)	−0.033 (0.022)	−0.009 (0.025)	−0.048* (0.027)	0.005 (0.032)
<i>App rating</i>	0.037** (0.017)	−0.023 (0.019)	0.007 (0.011)	0.001 (0.011)	0.040*** (0.009)	0.015 (0.009)	0.010 (0.007)	−0.022*** (0.007)	−0.005 (0.007)	−0.011* (0.006)
Observations	6,874,960	6,874,960	6,874,960	6,874,960	6,874,960	6,874,960	6,874,960	6,874,960	6,874,960	6,874,960

Notes. This table shows regressions of metrics on app ratings and dummies for having received notifications. Panel A uses a single dummy for having received notifications. Panel B uses different dummies for different stages in the notification process. All regressions include driver and trip characteristics fixed effects and control for a quadratic function of the number of Uber trips the driver has completed. Standard errors are clustered by driver. All driving metrics are normalized to mean of zero and variance of one.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

strategy to address these potential endogeneity concerns. Our findings are consistent with these results.

Panel B of Tables 4 and 5 separates the effect of notifications across different notification states. The state is defined by the last event that took place. All effects are measured relative to the level before getting any notification. As we can see, effects tend to have the same signs as the main effect, although not all coefficients are significant. Notification 2 does not seem to have any effect on top of the original effect of notification 1. Notification 3, on the other hand, seems to have the strongest effect, consistent with the imminent threat of deactivation.

The main takeaway from these results is that, once drivers enter the deactivation process and start receiving notifications, they make substantial improvements to their behavior. This is evidence that the notifications system is working as intended. Importantly, these effects are persistent, because they remain after notifications expire—after drivers start behaving better and receive enough high ratings to get out of the deactivation process.

4.2. Driver Deactivation

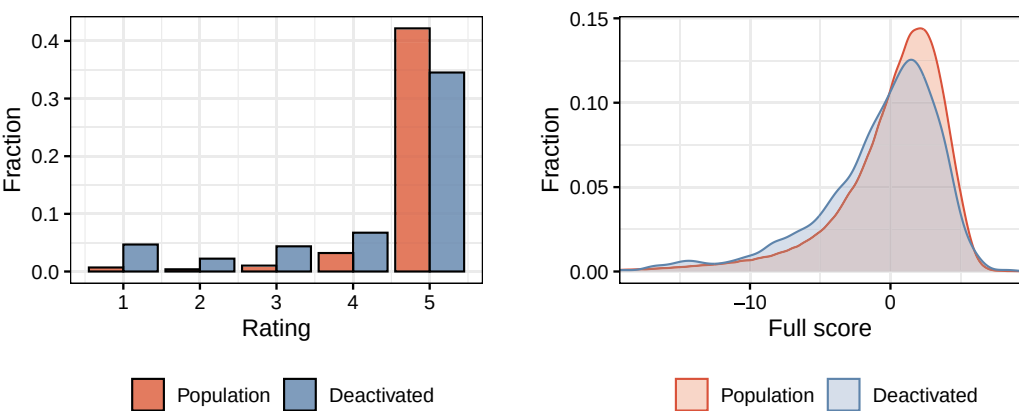
We now show evidence that the deactivation process described in the previous section is able to weed out

drivers that perform badly. Figure 6 compares the distribution of ratings and scores completed by drivers who were deactivated with the distribution for the general population. For deactivated drivers, we want to avoid looking at their behavior after responding to incentives, so we only consider trips that took place before they got their first warning.

Deactivated drivers have a much larger share of trips with low ratings. This relation is mechanical—Drivers are deactivated when they have low ratings. But we see that low scores are also more common among drivers that are eventually deactivated. In particular, we see that the distribution of scores has a longer left tail: The deactivation process disproportionately kicks out drivers that complete trips with very bad scores.

We now follow a more nuanced regressions analysis. We start, again, with the sample of trips of drivers who were eventually deactivated, but which took place before the driver got the first notification. To avoid confounders related to the time when trips took place or the places where trips took place, we match every such trip to the nearest ten trips by drivers who were not deactivated. We use a Euclidean metric in which a half-hour difference is equivalent to a difference in origin or destination of one kilometer. We then estimate the effect of

Figure 6. (Color online) Distribution of Ratings and Scores for Drivers Who Were Deactivated



Notes. These figures show the distribution of the rating and of the full score for two different samples: the whole population of trips and the trips by drivers who were eventually deactivated *that took place before getting the first notification*.

the trip being done by a driver who was eventually deactivated as

$$\hat{\tau} = \frac{1}{N} \sum_{i \in I_D} \left(y_i - \frac{1}{10} \sum_{j \in I_{ND,i}} y_j \right), \tag{5}$$

where y_i is the outcome variable for trip i , I_D is the set of trips that were completed by a driver who was eventually deactivated, and $I_{ND,i}$ denotes the set of i 's nearest neighbors among trips by drivers who were never deactivated.¹⁴ We compute standard errors as in Abadie and Imbens (2006) with an adjustment for clustering by driver.

Table 6 presents the results from this exercise. Drivers who are deactivated perform badly in terms of all four scores. They also tend to display worse behavior in terms of individual metrics. They are less likely to mount their phone and more likely to handle it. There is

no significant difference in terms of brakes and accelerations, but they are slow drivers, their trips are longer in terms of distance and duration, and they pick up and drop off riders farther away from the desired locations. All these results are consistent with the hypothesis that the deactivation process is able to eliminate drivers who had bad driving behavior, offsetting to some extent the fact that they could have been able to enter the platform because of simplified screening.

4.3. Information on Past Behavior

We now quantify the extent to which drivers' behavior responds to feedback provided by Uber on their past behavior.

Uber informed drivers of their past performance in terms of driving metrics.¹⁵ Every week, drivers received a simple report on their smartphone screen summarizing their driving metrics in the past week and

Table 6. Characteristics of Trips by Drivers Who Were Later Deactivated

	Dependent variable						
	Score F (1)	Score D (2)	Score R (3)	Score S (4)	Mounted (5)	Handling (6)	Brakes (7)
Deactivated	−0.914*** (0.137)	−0.471*** (0.164)	−0.925*** (0.114)	−0.123** (0.062)	−0.412*** (0.143)	0.442*** (0.162)	0.038 (0.063)
Observations	3,028	3,028	3,028	3,028	3,028	3,028	3,028

	Dependent variable						
	Accelerations (1)	Speed low (2)	Speed high (3)	Distance (4)	Duration (5)	Dropoff gap (6)	Pickup gap (7)
Deactivated	−0.041 (0.087)	−0.285*** (0.039)	−0.214*** (0.060)	0.080*** (0.022)	0.157*** (0.037)	0.049*** (0.019)	0.210*** (0.026)
Observations	3,028	3,028	3,028	3,028	3,028	3,028	3,028

Notes. This table shows regressions of driving metrics as well as scores on dummies for whether the driver was eventually deactivated. The sample includes all trips by riders who were not deactivated, and trips by drivers who were deactivated *before they got their first notification*. All metrics are normalized to mean of zero and variance of one. Standard errors are clustered by driver.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

comparing them with other drivers' metrics. Receiving this information could motivate drivers to improve their behavior, but it could also lead to worse behavior if well-behaving drivers decrease their efforts.

4.3.1. Experimental Setup. Every driver in our sample received the simple report, so there is no direct way to tell to what extent it affected their behavior. However, Uber ran a closely related experiment. Uber was considering a major upgrade of the simple report. The new version would be a complete dashboard in the Uber app with detailed information about past behavior. The dashboard included information on individual trips and on specific segments within each trip. Figure 7 shows screenshots of the dashboard.

Uber initially treated a random set of drivers with the upgraded dashboard and kept on sending the original report to a control group. We use this experimental setting to measure how additional information on past behavior influenced trip quality. In Online Appendix F.1, we report results of a balance test of pre-experiment averages by driver for all metrics and scores. We include only drivers who took at least 10 trips in the month before the experiment. We do not find statistically significant differences for most metrics and scores, but we find significant differences for the speed metrics and for the distance metric. Because we do not have perfect balance in the pre-treatment period for these variables, we control for pre-treatment averages in analyzing the experiment.

We start by analyzing the results of the experiments with regressions of the form

$$y_i = \alpha + \tau T_{d(i)} + \gamma X_i + \epsilon_i, \quad (6)$$

where y_i is some outcome and $T_{d(i)}$ is a dummy for whether the driver was in the treatment group. Our estimate for τ is thus an estimate of the intent-to-treat (ITT) effect. We limit our sample to those trips that took place

after the experiment started; X_i includes the driver's pre-experiment mean for the outcome and trip characteristics fixed effects.

We are also interested in measuring the average treatment effect (ATE) of interacting with the new dashboard. We construct an indicator variable I_i that is equal to one if driver $d(i)$ interacted with the dashboard in the week prior to trip i . We are then interested in a model of the form

$$y_i = \alpha + \theta I_i + \gamma X_i + \epsilon_i. \quad (7)$$

We estimate this regression by two-stage least squares (2SLS), instrumenting I_i with the treatment dummy and its interaction with the driver's pretreatment mean for each of the telemetry measures and scores. Treatment status and the pretreatment outcomes are demeaned for interpretability.¹⁶ Our estimate of θ is thus an estimate of the ATE.

Columns (1)–(4) in Table 7 show results for regressions of the form of Equations (6) and (7) where the dependent variables are our four scores. In panel A, we measure a positive effect of the ITT, although it is only significant for the full and routing scores. Panel B shows that interacting with the dashboard leads to a significant improvement in all four scores. The effect is especially pronounced for the full and control scores, where we would have expected the strongest impact. Interestingly, we also find an effect on the routing score which may be somewhat surprising given that the feedback provided is unrelated to routing. This might occur because feedback about their behavior may increase overall awareness of their behavior, even in dimensions not directly addressed by the feedback. We see the smallest effect on the speed score, possibly because the feedback (intended to encourage slower driving) is misaligned with rider preferences. Online Appendix F.2 presents a similar table for individual metrics. We do not measure significant effects.¹⁷

Figure 7. (Color online) Dashboard with Detailed Information About Past Driving Behavior

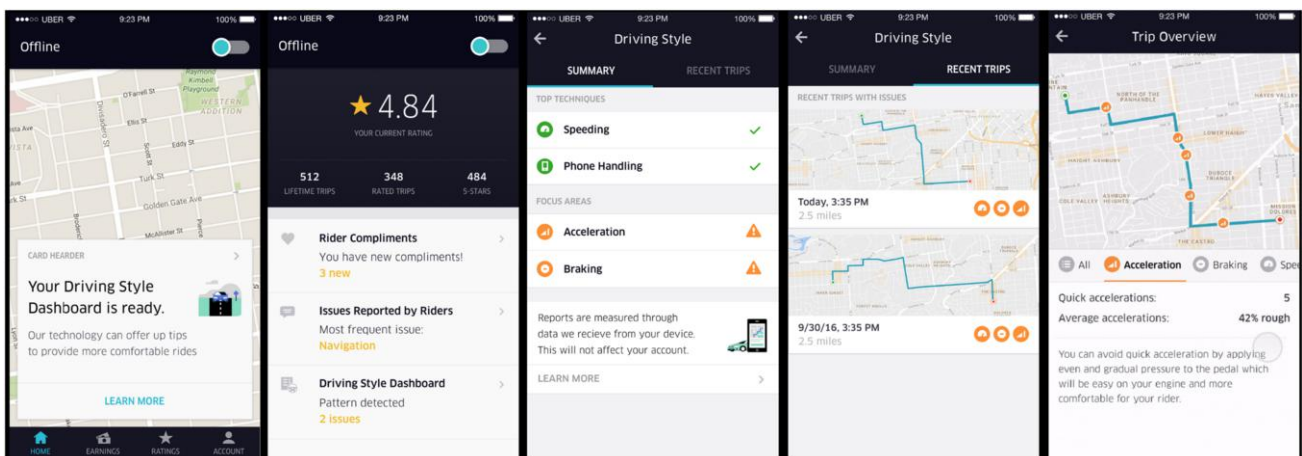


Table 7. Effect of the Driving Dashboard

	Dependent variable							
	Score F (1)	Score C (2)	Score R (3)	Score S (4)	Score F (5)	Score C (6)	Score R (7)	Score S (8)
Panel A: Intent to treat estimator								
Treatment	0.040*** (0.011)	0.018* (0.009)	0.028*** (0.008)	0.004 (0.004)				
Treatment × Not Bottom 10th Percentile					0.046*** (0.015)	0.010 (0.014)	0.032*** (0.011)	0.007 (0.005)
Treatment × Bottom 10th Percentile					0.106* (0.061)	0.171*** (0.058)	−0.012 (0.050)	0.043 (0.040)
Preperiod Mean	0.607*** (0.008)	0.820*** (0.006)	0.569*** (0.006)	0.767*** (0.011)				
Bottom 10th Percentile Before					−2.025*** (0.045)	−2.590*** (0.043)	−1.155*** (0.039)	−0.752*** (0.033)
Observations	3,786,006	3,786,006	3,786,006	3,786,006	3,786,006	3,786,006	3,786,006	3,786,006
Panel B: 2SLS estimator								
Interaction	0.069*** (0.019)	0.051*** (0.017)	0.032** (0.015)	0.016** (0.008)				
App Interaction × Not Bottom 10th Percentile					0.059*** (0.019)	0.027 (0.017)	0.036** (0.015)	0.015** (0.007)
App Interaction × Bottom 10th Percentile					0.206** (0.103)	0.311*** (0.097)	−0.033 (0.081)	0.031 (0.040)
Bottom 10th Percentile Before					−0.016 (0.042)	−0.049 (0.047)	−0.003 (0.035)	−0.016 (0.019)
Observations	3,786,006	3,786,006	3,786,006	3,786,006	3,786,006	3,786,006	3,786,006	3,786,006

Notes. Panel A shows regressions of scores on a dummy for being in the treatment group of the dashboard experiment and on the preperiod mean score. Panel B shows 2SLS regressions of scores on a dummy for observing the dashboard, using the treatment dummy and interactions with pretreatment metrics and scores as instruments. Columns (4)–(6) measures heterogeneous effects, depending on whether the driver was among the 10% of worst performing drivers in terms of the pretreatment mean of the outcome variable. Standard errors are clustered by driver.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

We are also interested in seeing how the effects of the dashboard differ by drivers’ pretreatment behavior. To see how the experiment affected poorly performing drivers, we first compute the average of each outcome for each driver in the pretreatment period (excluding drivers with fewer than 10 trips in the month before the experiment launched). We then code a pretreatment dummy variable, *Bottom 10th Percentile Before*, which indicates whether the driver was among the worst-performing 10% of drivers in terms of the outcome variable during the preperiod.¹⁸

Columns (5)–(8) in Table 7 present results from regressions similar to Equations (6) and (7), where access and interaction with the dashboard are interacted with the pretreatment dummy. For the full and control scores, the effects of the dashboard are primarily driven by drivers who performed worst during the pretreatment period. These findings align with those of Choudhary et al. (2022), who observed similar results for general private drivers in India. This suggests that informing poorly performing drivers about their performance encourages increased efforts and leads to performance improvement. Additionally, there appears to be a small improvement for drivers who were not performing poorly, indicating that more detailed information

does not deteriorate the performance of previously well-performing drivers. The patterns for the routing and speed scores are less clear and noisier. This is consistent with our claim that these scores are less likely to be influenced by feedback, suggesting that one should not expect a larger improvement from the lowest-performing drivers.

4.4. Response to Individual Rider Effects

Our previous results in this section show how drivers improve their driving behavior in response to incentives and nudges from the platform. We now analyze whether individual riders have an impact on driving behavior. One takeaway from Table 2 is that rider effects account for a substantial fraction of the variation, especially for speed and routing metrics. Here we explore if drivers tailor their behavior to riders’ heterogeneous preferences. For instance, it could be the case that some individual riders tend to be in a hurry and put pressure on their driver to get to their destination quickly. Such findings would suggest that Uber’s rating systems encourage drivers to follow the form of driving riders want.

For some outcome y_i , let $\bar{y}_i^{LO}$ be the average value of the outcome for all trips taken by $r(i)$, the rider who took

trip i , leaving out the current trip. We run

$$y_i = \mu_{d(i)} + \nu_{c(i)} + \beta \bar{y}_i^{LO} + \epsilon_i, \quad (8)$$

where the first two terms denote driver and trip characteristics fixed effects. Our main object of interest is β , which measures to what extent y_i is correlated within riders. The driver and trip fixed effects account for the fact that riders will tend to have similar trips (e.g., trips from their home) and that drivers may focus their driving around certain areas. Many riders only have a small number of trips in our database, so $\bar{y}_i^{LO}$ could be a noisy measure. Thus, we restrict our sample to riders with 20 or more trips.¹⁹

Panel A in Table 8 shows results for this exercise, where the outcome variables are metrics. We see a negative but negligible effect on mounted and handling. On the other hand, we see a strong and positive effect on brakes and accelerations and especially on speed and routing metrics. This means that there is a strong correlation of metrics within riders. We interpret this finding as establishing that riders influence driver behavior. Online Appendix C.1 presents similar results for scores: The strong correlations in metrics are reflected as correlations in scores.

We also analyze whether the effect individual riders have on driving intensifies during rush hour, when riders are presumably in a hurry.²⁰ Online Appendix D.4 confirms that riders who take trips during the morning rush hour tend to prefer faster trips, based on regressions similar to those in Table 1 but that also allow for heterogeneous preferences. One should then expect stronger within-rider correlations during the morning rush hour if riders who use UberX at that time do so consistently.²¹ We estimate regressions of the form of Equation (8), but where we also interact $\bar{y}_i^{LO}$ with

dummies for different times of the week: morning rush hour, afternoon rush hour, and off-peak times. Panel B in Table 8 show the result of this exercise. We see especially large coefficients for trips during the morning rush hour, as expected.

5. UberX vs. UberTaxi

Having studied various dimensions in which UberX drivers respond to Uber's nudges and incentives, we now compare the driving behavior of UberX and UberTaxi trips. The differences we measure cannot be attributed solely to the different quality control mechanisms—ratings, information, and nudges for UberX and ex ante screening for UberTaxi—because many other factors vary between the two products. The demographics of both pools of drivers differ, for instance, and the drivers face different monetary incentives. However, our estimates measure the causal effect of requesting an UberX instead of an UberTaxi for the same trip. Beyond its direct relevance for a rider deciding which product to choose, this causal effect tells us whether a ride-hailing platform that relies almost entirely on ex post quality control can achieve the same level of quality of one that relies on ex ante screening.

The main question we address is as follows: Given the characteristics (origin, destination, and time) of an UberTaxi trip, how would the driving behavior experienced by the rider have changed if she had instead requested an UberX trip? In other words, we aim to estimate the average treatment effect of getting an UberX trip, for the distribution of UberTaxi trips. This estimate is, hence, the average treatment effect on the untreated (ATU). We do not estimate the more conventional average treatment effect on the treated (ATT) because, although we can find similar UberX trips for most

Table 8. Response to Rider Preferences

	Dependent variable									
	Mounted (1)	Handling (2)	Brakes (3)	Accelerations (4)	Speed low (5)	Speed high (6)	Distance (7)	Duration (8)	Pickup (9)	Dropoff (10)
Panel A: Main effect										
Mean by rider	−0.006*** (0.001)	−0.004 (0.002)	0.140*** (0.003)	0.113*** (0.003)	0.338*** (0.002)	0.352*** (0.003)	0.518*** (0.003)	0.516*** (0.003)	0.545*** (0.002)	0.601*** (0.002)
Observations	3,231,440	3,231,440	3,231,440	3,231,440	3,231,440	3,231,440	3,231,440	3,231,440	3,231,440	3,231,440
Panel B: Heterogeneous effects by time of the week										
Mean by rider	−0.002 (0.002)	0.001 (0.003)	0.158*** (0.003)	0.125*** (0.003)	0.302*** (0.003)	0.326*** (0.003)	0.432*** (0.004)	0.457*** (0.003)	0.502*** (0.003)	0.525*** (0.003)
× Off-peak										
Mean by rider	−0.025*** (0.004)	−0.028*** (0.006)	0.110*** (0.007)	0.101*** (0.007)	0.512*** (0.006)	0.507*** (0.008)	0.861*** (0.007)	0.752*** (0.006)	0.704*** (0.005)	0.974*** (0.005)
× AM rush										
Mean by rider	−0.006* (0.003)	−0.002 (0.005)	0.104*** (0.006)	0.083*** (0.006)	0.310*** (0.005)	0.311*** (0.006)	0.428*** (0.006)	0.463*** (0.006)	0.531*** (0.005)	0.403*** (0.004)
× PM rush										
Observations	3,231,440	3,231,440	3,231,440	3,231,440	3,231,440	3,231,440	3,231,440	3,231,440	3,231,440	3,231,440

Notes. This table shows regressions of quality metrics on leave-out means by rider. Only riders with more than 20 trips are included. All regressions control for driver and trip characteristics fixed effects. Standard errors are clustered by driver.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

UberTaxi trips, the reverse is not true due to the much smaller number of UberTaxi trips in our data. The difference between an ATU and ATT is not particularly relevant in this setting because what is a treatment and what is a control is arbitrary: We could have just as easily defined the treatment to be requesting an UberTaxi trip, in which case the estimate we obtain is the negative of the ATT. The key assumption underlying our estimation is unconfoundedness when controlling for trip characteristics: Conditional on the trip's origin, destination, and time of day, there are no further unobserved characteristics of trips that would impact the telemetry metrics.

We use two different methodologies with very similar results. First, similar to the matching estimator we describe in Equation (5), we match UberX and UberTaxi trips according to their origin and destination coordinates and the hour of the week at which they started. Concretely, we match every UberTaxi trip to its 10 nearest UberX neighbors, using a Euclidean metric in which a half-hour difference is equivalent to a difference in origin or destination of 1 kilometer. We then estimate the UberX effect as

$$\hat{\tau}^{match} = \frac{1}{N} \sum_{i \in I_{taxi}} \left(y_i - \frac{1}{10} \sum_{j \in C_i} y_j \right), \quad (9)$$

where I_{taxi} is the set of UberTaxi trips, and C_i denotes the set of i 's nearest neighbors among UberX trips. We compute standard errors as in Abadie and Imbens (2006) with an adjustment for clustering by driver.

The second methodology is a simple fixed effects regression using the same trip characteristics as in Section 3. The specification we run is

$$y_i = \tau x_i + \beta X_i + \epsilon_i, \quad (10)$$

where x_i is a dummy that equals one if driver $d(i)$ is an UberX driver, and βX_i is a set of fixed effects. Our coefficient of interest is τ . In general, this estimator is not consistent for the ATU as it is usually defined, but it converges to a weighted average of treatment effects. As we see below, both methodologies result in almost identical results.

Table 9 shows the estimates from these specifications for metrics as the dependent variable. The matching estimator (panel A) has almost identical results to the ordinary least squares (OLS) estimator with trip characteristics fixed effects (panel B). Our estimates change somewhat if we also include rider fixed effects (panel C) but not enough for the interpretation of the coefficients to change.

UberX drivers are more likely to mount their cell phones. The difference is almost one standard deviation. This is not surprising: Uber has led campaigns to ensure drivers mount their phones, sometimes giving away phone mounts. UberX drivers are, however, also more likely to handle their phones. This is also not too

surprising because they rely more on their cell phone to find the next trip, and they are probably more tech-savvy and thus more likely to navigate with their phones.

UberX drivers have fewer hard brakes and accelerations. The difference is especially pronounced for accelerations, with a difference of roughly one quarter of a standard deviation. They also drive more slowly in terms of both speed measures. However, the UberX effect is much stronger for *speed high* than for *speed low*, which means that UberX drivers tend to drive at a steadier speed than UberTaxi drivers, as one would expect if they pay more attention to riders' preference for steady speeds.

UberX drivers perform worse in terms of both the distance and duration of trips.²² This finding aligns with Liu et al. (2021), who find that UberX drivers in New York generally take longer routes than taxi drivers. Online Appendix C.3 shows that UberX drivers have more trips that are longer than Uber's expected route, which may indicate moral hazard—driving longer to increase earnings. However, Online Appendix C.3 also shows that UberX drivers have fewer trips that are shorter than Uber's expected route, suggesting that they are less likely to deviate from the suggested route to take shortcuts, possibly due to less experience with the city's streets. Finally, UberX drivers tend to pick up and drop off riders closer to their desired locations, which could also contribute to the longer trips that we observe.

Table 10 reports estimates of the ATE on driving scores. The results reveal a mixed picture. UberX drivers perform better in terms of the control score, particularly in dimensions related to speed. However, they perform worse in terms of the routing score. The full score, which aggregates all dimensions, suggest that UberX and UberTaxi drivers perform roughly equally, with perhaps a moderate advantage for UberX drivers.

The main takeaway from this analysis is that there is no clear answer as to whether UberX or UberTaxi drivers provide a better quality in terms of the metrics we analyze. On one hand, UberX drivers perform better in terms of braking, accelerating, and driving at appropriate speeds. On the other hand, they perform worse in terms of handling their phones and routing. Overall, our findings suggest that, on average, passengers either perceive little difference between the overall driving quality of the two types of drivers or tend to slightly prefer UberX drivers.

In the Online Appendix, we consider a variety of extensions and robustness checks; for example, in Online Appendix D.5, we make use of estimates of the market value of the cars and show that our results are robust to comparing UberX and UberTaxi rides with similar market prices. One caveat is that, as we show in Online Appendix B.4, some of our results suggest that UberTaxi riders may prefer the driving behavior of

Table 9. Comparison of Driving Behavior Between UberX and UberTaxi Trips

	Dependent variable									
	Mounted (1)	Handling (2)	Brakes (3)	Accelerations (4)	Speed low (5)	Speed high (6)	Distance (7)	Duration (8)	Pickup (9)	Dropoff (10)
Panel A: Matching estimator										
UberX	0.9541*** (0.0247)	0.0743*** (0.0129)	−0.0481*** (0.0119)	−0.2524*** (0.0144)	−0.0174** (0.0085)	−0.2172*** (0.0071)	0.2048*** (0.0032)	0.1681*** (0.0058)	−0.0824*** (0.0039)	−0.2188*** (0.0066)
Observations	139,669	139,669	139,669	139,669	139,669	139,669	139,669	139,669	139,669	139,669
Panel B: Trip characteristics fixed effects										
UberX	0.9517*** (0.0250)	0.0758*** (0.0138)	−0.0360*** (0.0121)	−0.2449*** (0.0147)	−0.0308*** (0.0087)	−0.2195*** (0.0073)	0.1982*** (0.0035)	0.1542*** (0.0061)	−0.0805*** (0.0040)	−0.2228*** (0.0066)
Observations	7,147,732	7,147,732	7,147,732	7,147,732	7,147,732	7,147,732	7,147,732	7,147,732	7,147,732	7,147,732
Panel C: Trip characteristics and rider fixed effects										
UberX	0.9471*** (0.0262)	0.0632*** (0.0148)	−0.0574*** (0.0133)	−0.2559*** (0.0159)	0.0019 (0.0101)	−0.1999*** (0.0089)	0.1526*** (0.0057)	0.1193*** (0.0083)	−0.0779*** (0.0064)	−0.2027*** (0.0081)
Observations	7,147,732	7,147,732	7,147,732	7,147,732	7,147,732	7,147,732	7,147,732	7,147,732	7,147,732	7,147,732

Notes. This table compares the driving behavior of UberX and UberTaxi trips according to driving metrics. Panel A uses a matching estimator, where every UberTaxi trip is compared with the ten nearest UberX trips based on origin and destination coordinates and time of the week. Panel B presents results from a linear regression that includes trip characteristics fixed effects. Panel C presents results from a linear regression that includes trip characteristics as well as rider fixed effects. All driving metrics are normalized to mean zero and variance one. Standard errors are adjusted as in Abadie and Imbens (2006) and clustered by driver.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

UberTaxi drivers. Those results, however, only account for the preferences of a sample of riders who constitute 2% of the trips in our data. An overwhelming majority of Uber riders in fact prefer the driving behavior of UberX drivers.

5.1. UberX vs. UberTaxi During Rush Hour

We now perform an exercise that gives suggestive evidence for whether UberX or UberTaxi drivers are more responsive to riders' preferences. As one might expect, we show in Online Appendix D.4 that riders prefer faster trips during rush hours. We thus analyze whether UberX trips are especially fast during those hours. To that end, we split our data set into morning rush hour trips, afternoon rush hour trips, and off-peak trips. We match every UberTaxi trip to its 10 nearest UberX trips within its subsample and compute the average treatment effect for each subsample, using the matching estimator from Equation (9).

Table 11 shows the result from this exercise. UberX drivers handle their phones more relative to UberTaxi drivers during rush hour than during off-peak hours. The gap in hard accelerations, hard brakes, and speed high shrinks during rush hour. Whereas speed low is lower for UberX drivers during off-peak hours, it is higher during rush hours. Furthermore, UberX drivers tend to take longer routes during rush hour but their trips are faster, and they tend to pick up and drop off riders closer to the desired locations during the morning rush hour. This behavior is consistent with UberX drivers being more responsive to passenger preferences, paying more attention to riders who want to get to their

destination on time by driving faster (therefore braking and accelerating more), by handling the phone more to find better routes to avoid traffic, and by rerouting to ensure the rider gets to the destination more quickly.²³ It is perhaps surprising that we also find similar (but

Table 10. Comparison of Driving Behavior Between UberX and UberTaxi Trips

	Dependent variable			
	Score F (1)	Score C (2)	Score R (3)	Score S (4)
Panel A: Matching estimator				
UberX	0.0324 (0.0242)	0.1184*** (0.0257)	−0.2866*** (0.0186)	0.2970*** (0.0092)
Observations	139,669	139,669	139,669	139,669
Panel B: Trip characteristics fixed effects				
UberX	0.0580** (0.0252)	0.1006*** (0.0265)	−0.2589*** (0.0197)	0.2869*** (0.0095)
Observations	7,147,732	7,147,732	7,147,732	7,147,732
Panel C: Trip characteristics and rider fixed effects				
UberX	0.1718*** (0.0337)	0.1425*** (0.0288)	−0.1504*** (0.0287)	0.2935*** (0.0121)
Observations	7,147,732	7,147,732	7,147,732	7,147,732

Notes. This table compares the driving behavior of UberX and UberTaxi trips according to driving scores. Panel A uses a matching estimator, where every UberTaxi trip is compared with the ten nearest UberX trips based on origin and destination coordinates and time of the week. Panel B presents results from a linear regression that includes trip characteristics fixed effects. Panel C presents results from a linear regression that includes trip characteristics as well as rider fixed effects. All driving metrics are normalized to mean zero and variance one. Standard errors are adjusted as in Abadie and Imbens (2006) and clustered by driver.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 11. Heterogeneity in Effect of UberX Using a Matching Estimator

	Dependent variable									
	Mounted (1)	Handling (2)	Brakes (3)	Accelerations (4)	Speed low (5)	Speed high (6)	Distance (7)	Duration (8)	Pickup (9)	Dropoff (10)
UberX (Off-peak)	0.949*** (0.025)	0.062*** (0.014)	−0.066*** (0.013)	−0.282*** (0.015)	−0.054*** (0.009)	−0.266*** (0.008)	0.202*** (0.004)	0.179*** (0.006)	−0.077*** (0.004)	−0.210*** (0.007)
UberX (AM rush)	0.989*** (0.032)	0.095*** (0.015)	−0.017 (0.016)	−0.221*** (0.018)	0.025** (0.011)	−0.160*** (0.009)	0.230*** (0.005)	0.164*** (0.008)	−0.103*** (0.007)	−0.238*** (0.009)
UberX (PM rush)	0.920*** (0.032)	0.081*** (0.016)	−0.041*** (0.014)	−0.209*** (0.018)	0.028** (0.013)	−0.158*** (0.009)	0.175*** (0.006)	0.139*** (0.010)	−0.069*** (0.008)	−0.217*** (0.011)
Observations	139,669	139,669	139,669	139,669	139,669	139,669	139,669	139,669	139,669	139,669

Notes. This table show the effect of UberX varies during rush hours. In order to do so, we split the sample into three subsamples: off-peak trips, morning rush hour trips, and afternoon rush hour trips. Each column presents results for a different response variable. We use a matching estimator, where every UberTaxi trip is compared with the 10 nearest UberX trips within the same subsample, based on origin and destination coordinates and time of the week. All driving metrics are normalized to mean of zero and variance of one. Standard errors are clustered by driver.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

smaller) effects during the afternoon rush hour, when riders are not in as much of a hurry as in the morning.

Online Appendix C.2 presents similar results for scores. The full and control scores are lower for UberX during the morning rush hour, whereas the routing score is higher. However, these results do not necessarily reflect whether these adjustments are good or bad for drivers; as shown in Online Appendix D.4, riders' preferences are somewhat different during the morning rush hour.

6. Conclusions

Observers have expressed concern that ride-hailing platforms might reduce quality by allowing inexperienced drivers on their platform, a consequence of their streamlined screening process that is designed to allow for a flexible workforce. Using objective measures of driving quality, we provide empirical evidence that the ratings, incentives, nudges, and information systems set up by Uber constitute important offsetting forces that significantly improve the quality of driving. Ultimately, we find that UberX drivers provide quality that is roughly comparable to that provided by UberTaxi drivers. Although we cannot fully attribute differences in quality to the differences in quality control mechanisms, our findings suggest that concerns about low quality on ride-hailing platforms are misguided. Thus, our findings indicate that policymakers concerned about quality should consider allowing marketplaces to rely on well-designed ex post quality control systems rather than on ex ante screening and occupational licensing.

Our paper highlights the need to understand the specific channels that contribute to the effectiveness of these quality control mechanisms, as well as to the differences between UberX and UberTaxi driving behavior. This question may be best answered by randomized experiments that can be conducted in the future. For instance, future research could investigate whether UberX drivers are primarily motivated by an intrinsic desire to

create a good experience for passengers or whether perceived or real economic incentives play a more important role in motivating behavior.

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Endnotes

- ¹ Because our scores are derived from UberX trips (which account for 98% of our data), the quality should be interpreted as reflecting the preferences of UberX riders; unfortunately our UberTaxi data are insufficient to reliably compare preferences across categories of riders.
- ² Jin and Leslie (2003) analyze providers' response to the regulation that existed before digital markets. Mayzlin et al. (2014) and Luca and Zervas (2016) analyze a different kind of provider behavior: They find evidence that hotel and restaurant owners post positive reviews for themselves and negative reviews for competitors.
- ³ The driver must at minimum be 21 years of age, possess an active, permanent driver's license in good standing, pass a national background check, pass a two-week public chauffeur course and licensing exam, have an authorized debt clearance or payment plan, and be in good standing with court-order child support payments.
- ⁴ Telemetry data may be less accurate when the phone is not mounted, although this problem is mitigated by using telemetry from the GPS instead of accelerometers.
- ⁵ An acceleration event takes place when speed increases during two consecutive two-second intervals. A braking event is defined similarly.
- ⁶ We exclude from our sample trips in which the pickup metric is greater than 150 meters or the dropoff metric is greater than 300 meters, because these are likely trips in which the rider decided to change the origin or destination. We also exclude trips in which the absolute value of the distance metric is greater than 0.7

or in which the absolute value of the duration metric is greater than 0.8.

⁷ In some specifications we also include rider fixed effects (see Online Appendix D.1). Although our numbers change somewhat, the interpretation of our results does not change. Another potential concern is that the quality of the car might alter the perception that a trip is safe. Online Appendix D.5 shows that car quality does not interact in meaningful ways with our safety variables.

⁸ We include higher order terms of speed variables because we expect them to have more important nonlinearities than the other four variables, which are defined as fractions. Fully interacting both terms would result in a regression with 291,600 terms, and it would not be feasible to estimate it given our sample size. We thus separate the function additively into three terms, one for phone usage and the second one for driving.

⁹ The region with highest density is above and to the left of the bliss point. The gradient at that point is consistent with the coefficients for speed high and speed low in Table 1.

¹⁰ In an experiment where both surrogates and the final outcome are observed for each unit, Athey et al. (2019) show that efficiency can be gained by pooling data from the treatment group and the control group when estimating the relationship between the surrogates and the final outcome. Our analysis of a randomized experiment uses a smaller dataset; in that case, efficiency is gained by using a larger dataset to estimate the relationship between surrogates and the final outcome.

¹¹ One challenge is that $\mu_{d(i)}$ captures most of the variation for drivers with few trips, and $\gamma_{r(i)}$ captures most of the variation for riders with few trips. We thus limit our sample to drivers and riders that have more than 20 trips in our sample.

¹² This finding underscores the importance of checking that our results about the comparison between UberX and UberTaxi are present even when considering metrics other than speed because results based on speed may rely more heavily on carefully controlling for trip characteristics.

¹³ The thresholds to go back are 4.6, 4.5, and 4.45 after notification 1, 2, and 3, respectively.

¹⁴ This estimator is an average treatment effect on the treated (ATT) because it weights observations according to the distribution of characteristics of trips by drivers that were eventually deactivated.

¹⁵ This feature of the product was discontinued in late 2018.

¹⁶ We obtain similar results if we use a less rich set of instrumental variables.

¹⁷ We pulled data from other cities to try to increase the power of our regressions, but we found inconsistent results on metrics. For scores, we found a stronger effect in San Francisco, somewhat weaker results in Los Angeles, Washington DC, and Boston, and no evidence of an effect in New York.

¹⁸ We obtain similar results if we create similar dummies with different cutoffs or if we use a continuous measure instead of a dummy variable.

¹⁹ Restricting to riders with 20 or more trips reduces the sample size in the regression. As a result, a higher share of trip characteristics groups have few or no trips. Accordingly we use coarser fixed effects than in previous regressions. We divide the sample into 64 groups by origin, 64 groups by destination, and 15 groups by hour of the week. See Online Appendix A for more details.

²⁰ We define morning rush hour trips as those starting during weekdays between 6 am and 10 am and afternoon rush hour trips as those starting between 5 pm and 9 pm on weekdays.

²¹ We note that that this coefficient might also pick up unobserved trip characteristics; one set of heavy UberX riders might be riders

who regularly use UberX during the morning commute, and the route may be identical, inducing correlation among scores within a rider.

²² For these results, we use the distance and duration metrics based on imputed expected duration and distance, as explained in Online Appendix B.3.

²³ Note, however, that this result could be driven by alternative explanations. For instance, it could be that UberX drivers are more likely to be in a hurry during rush hours and thus drive faster.

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